5	(a)	Explain how line spectra, together with the concept of photon, provide evidence for discrete energy levels in atoms.	
		.....	
		.....	
		.....	[2]
	L1	**For each emitted photon, the energy is equal to the <u>difference</u> between the energy levels in the atom across which de-excitation of the atom occurred.** **The line spectra only show photons of certain allowed energies and this proves that the energies in an atom can only possess discrete and not continuous values.**	**B1** **B1**
	(b)	A discharge lamp contains excited atoms which emit electromagnetic radiation at several wavelengths. Figure 5.1 shows the three lowest electron energy levels of one of these atoms, labeled n = 1, 2 and 3.	

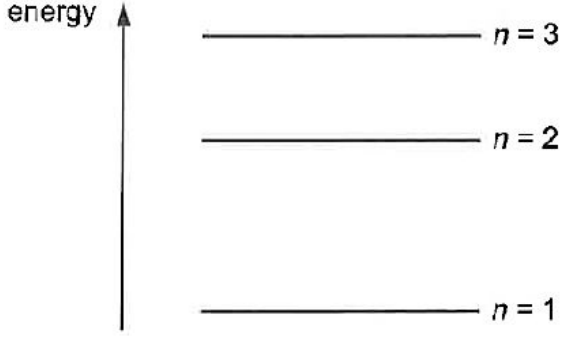


Figure 5.1

Electron transitions between these three energy levels can produce three different wavelengths of radiation. The transition from n = 2 to n = 1 produces radiation of wavelength 440 nm. Radiation of wavelength 590 nm is also emitted.

		(i)	Show that wavelength of the radiation emitted from the third transition is 250 nm.

				[2]
		L2	$E_{photon} = \Delta E$ $\frac{hc}{\lambda} = \Delta E$ **590 nm photon** must be emitted from the **smallest** energy transition ***n* = 3 to *n* = 2.** The **third transition** must therefore be from ***n* = 3 to *n* = 1**. Let the wavelength of the photon emitted from the third transition be λ_3. $\boldsymbol{\Delta E_{3\to1} = \Delta E_{3\to2} + \Delta E_{2\to1}}$ $\boldsymbol{\frac{hc}{\lambda_3} = \frac{hc}{590\text{ nm}} + \frac{hc}{440\text{ nm}}}$ $\boldsymbol{\frac{1}{\lambda_3} = \frac{1}{590\text{ nm}} + \frac{1}{440\text{ nm}}}$ $\boldsymbol{\lambda_3 = 252\text{ nm} \approx 250\text{ nm}}$ *Notes:* $E_{photon} = \Delta E$ $\frac{hc}{\lambda} = \Delta E$ *The longer the wavelength λ, the smaller the electron energy transition in ΔE the atom.* *Only 3 energy transitions are possible among these 3 energy levels. Thus, since the 440 nm photon is emitted from the transition n = 2 to n = 1, the 590 nm photon must be emitted from the transition n = 3 to n = 2. The third energy transition must therefore be from n = 3 to n = 1.* *While the sum of energies of the photons from the first two transitions should be equal to the energy of the photon from the third energy transition, the wavelengths do not add up because* $\lambda \propto \frac{1}{\Delta E}$ *.*	**B1** **M1** **A0**
		(ii)	State the region of the electromagnetic spectrum in which the third wavelength occurs.	
				[1]

			.....	
		L2	**Ultraviolet** ***Do not accept UV or wrong spelling.***	**B1**

[Total: 5]